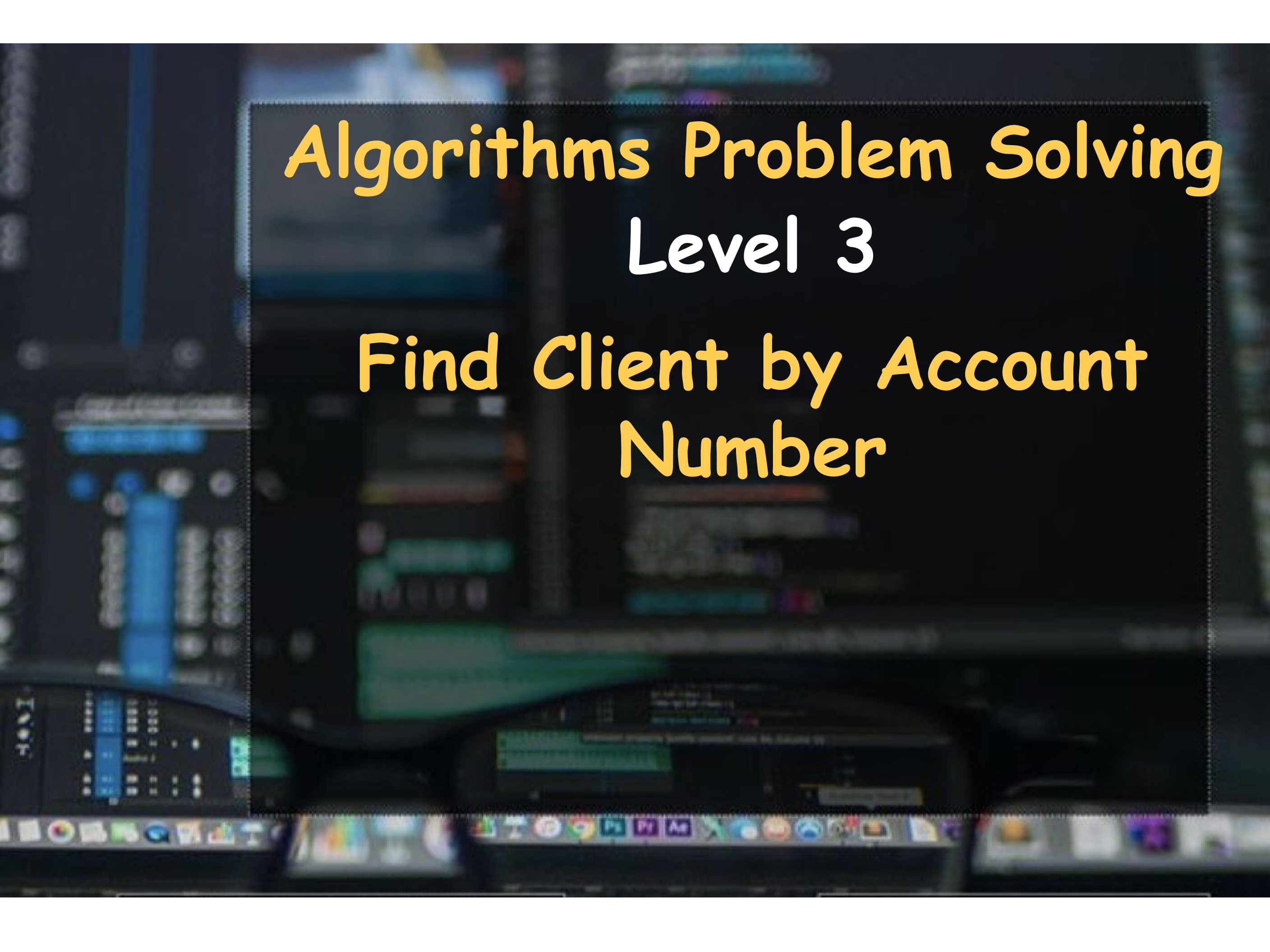




Algorithms Problem Solving

Level 3

Find Client by Account Number



Algorithms Problem Solving

Level 3

Find Client by Account Number

Definition

Key-Based Client Search

-  Program searches stored client records using `AccountNumber` as the lookup key.

-  When a matching record is found, its complete client data is returned through a reference parameter.

Explanation

Data Loading

-  The client file is loaded into a ``vector<sClient>``.
-  This separates file access from the actual search operation.

Linear Search

-  The vector is traversed from the first client to the last.
-  Each ``AccountNumber`` is compared with the requested account number.

Reference Result

-  When a match is found, the corresponding client structure is copied into the output reference parameter.
-  The function returns ``true`` to indicate success.

Not-Found Case

-  If no account matches, the function returns ``false``.
-  The caller can then display an appropriate "Not Found" message.

Procedures

1. Read the target Account Number.
2. Load client records from the file.
3. Traverse the vector.
4. Compare each Account Number with the target.
5. Copy the matching client into the output parameter.
6. Return `true` if found.
7. Return `false` otherwise.

💡 Example

Searching for:

A150

➡ Result

Client Details:

Account Number: A150

Name: Mohammed

Phone: 079

Balance: 5000

Note

- ◆ Account Number acts as the lookup key in this exercise.
- ◆ A linear search has $O(N)$ worst-case time complexity for N clients.
- ◆ If the records were indexed by account number, lookup could be significantly faster.

⊘ Warning

- ◆ The output reference should only be used as a valid result when the function returns ``true``.
- ◆ Account-number comparison should follow a consistent case convention.
- ◆ Searching the file repeatedly for every request can become inefficient for large datasets.



Comparison



`FindClientByAccountNumber()`

- ✓ Searches for one specific key.
- ✓ Returns a boolean success status.
- ✓ Provides the matching client through a reference parameter.



`PrintAllClients()`

- ✓ Traverses every client.
- ✓ Displays all records regardless of the requested key.
- ✓ Does not perform targeted filtering.

✦✦ Summary

- ✓ Loads client records into memory.
- ✓ Searches by unique Account Number.
- ✓ Uses linear comparison to locate the target.
- ✓ Returns both a success status and the matching client data.

Thank you 😊